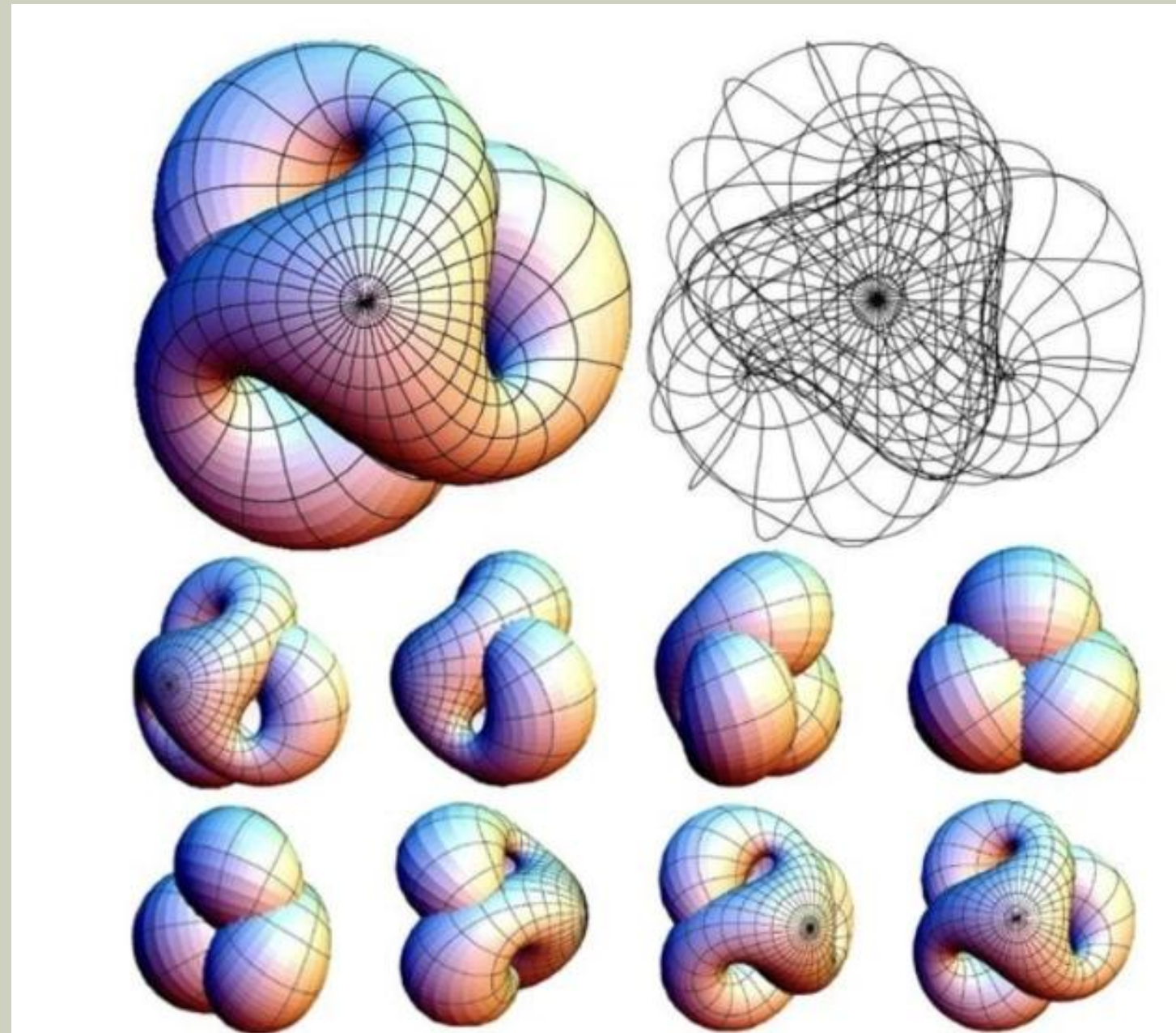




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TOPOLOGICAL DATA ANALYSIS FOR FINANCIAL MARKETS

By Darren Zheng

PROBLEM

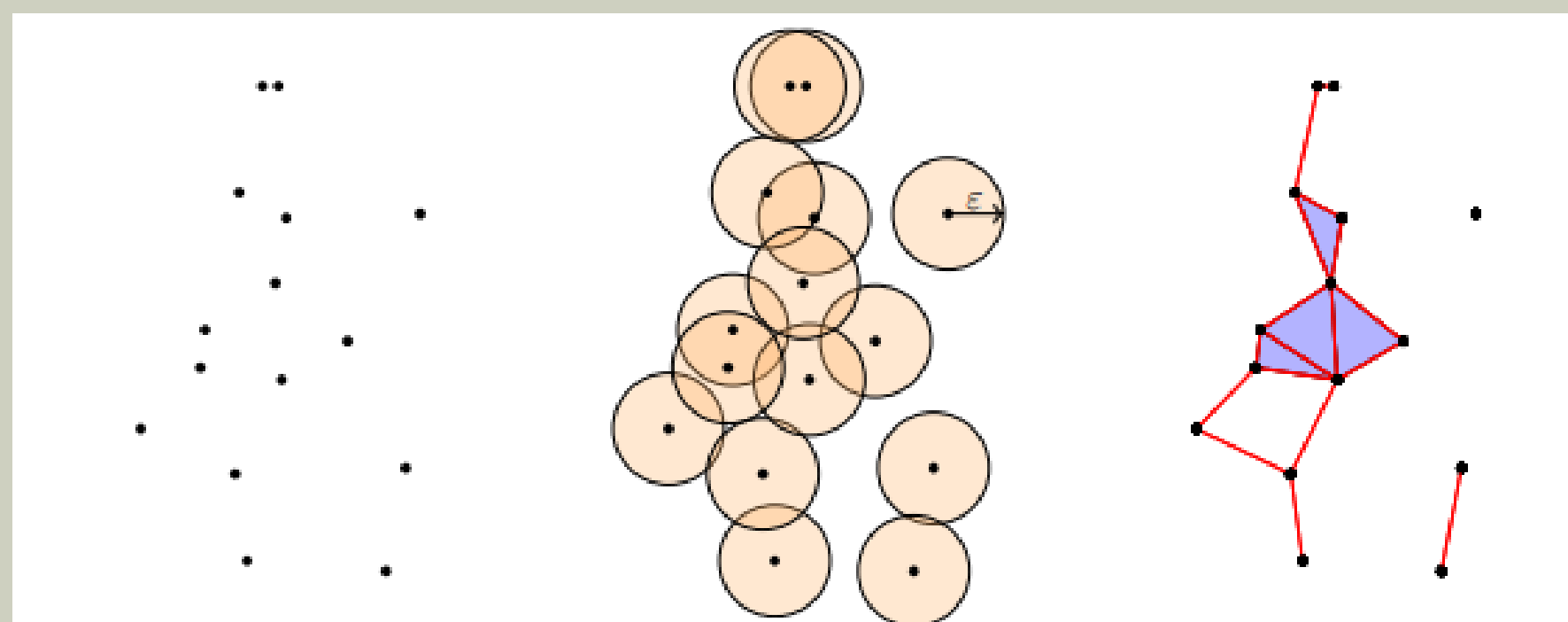
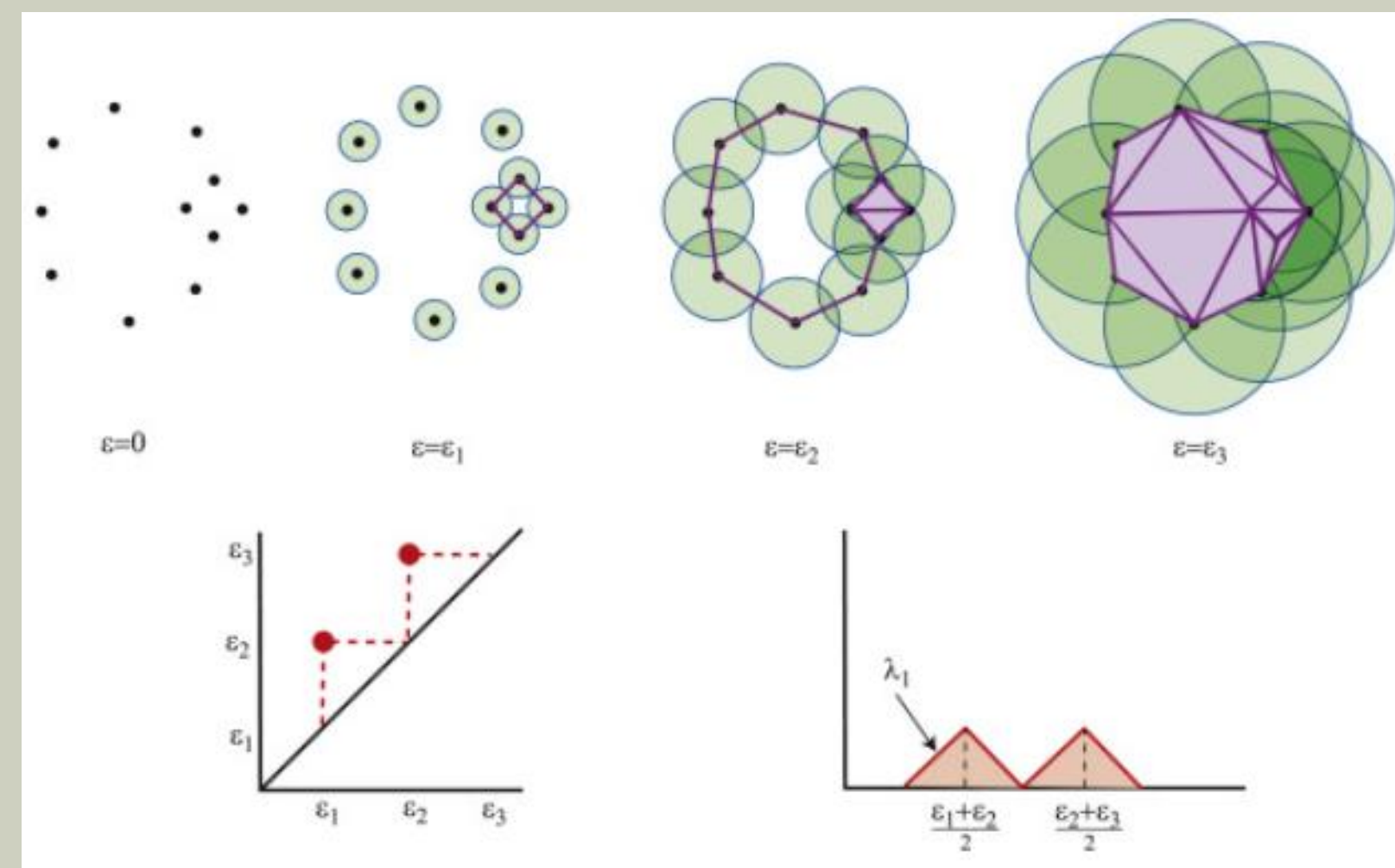
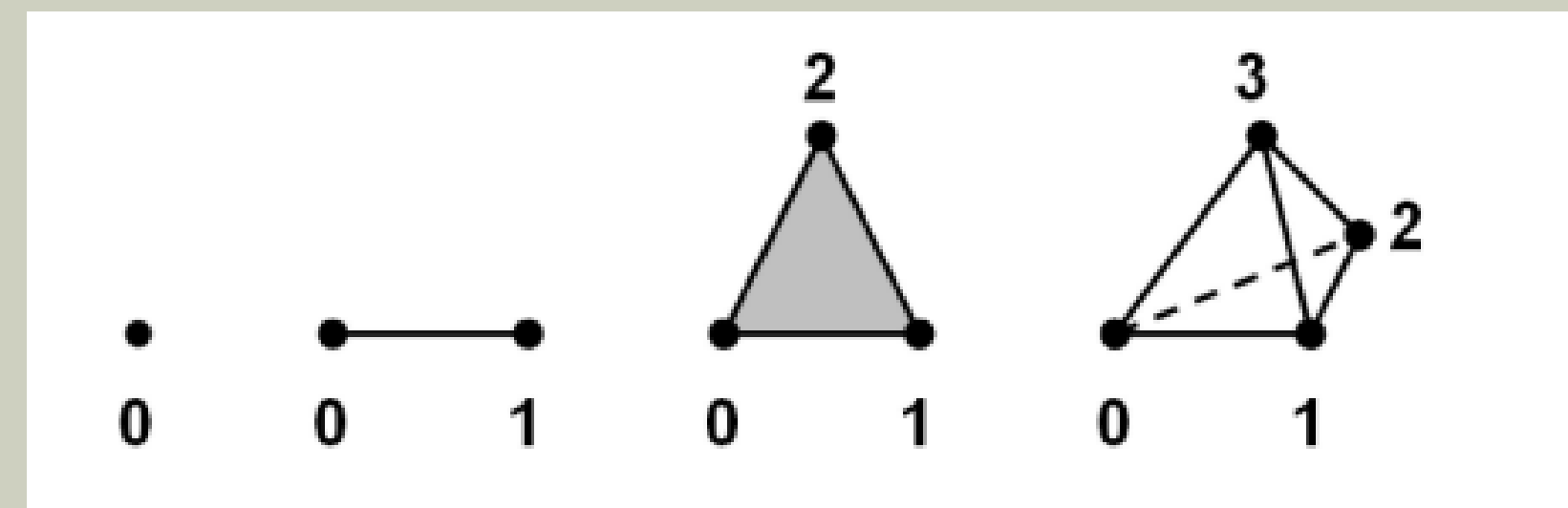
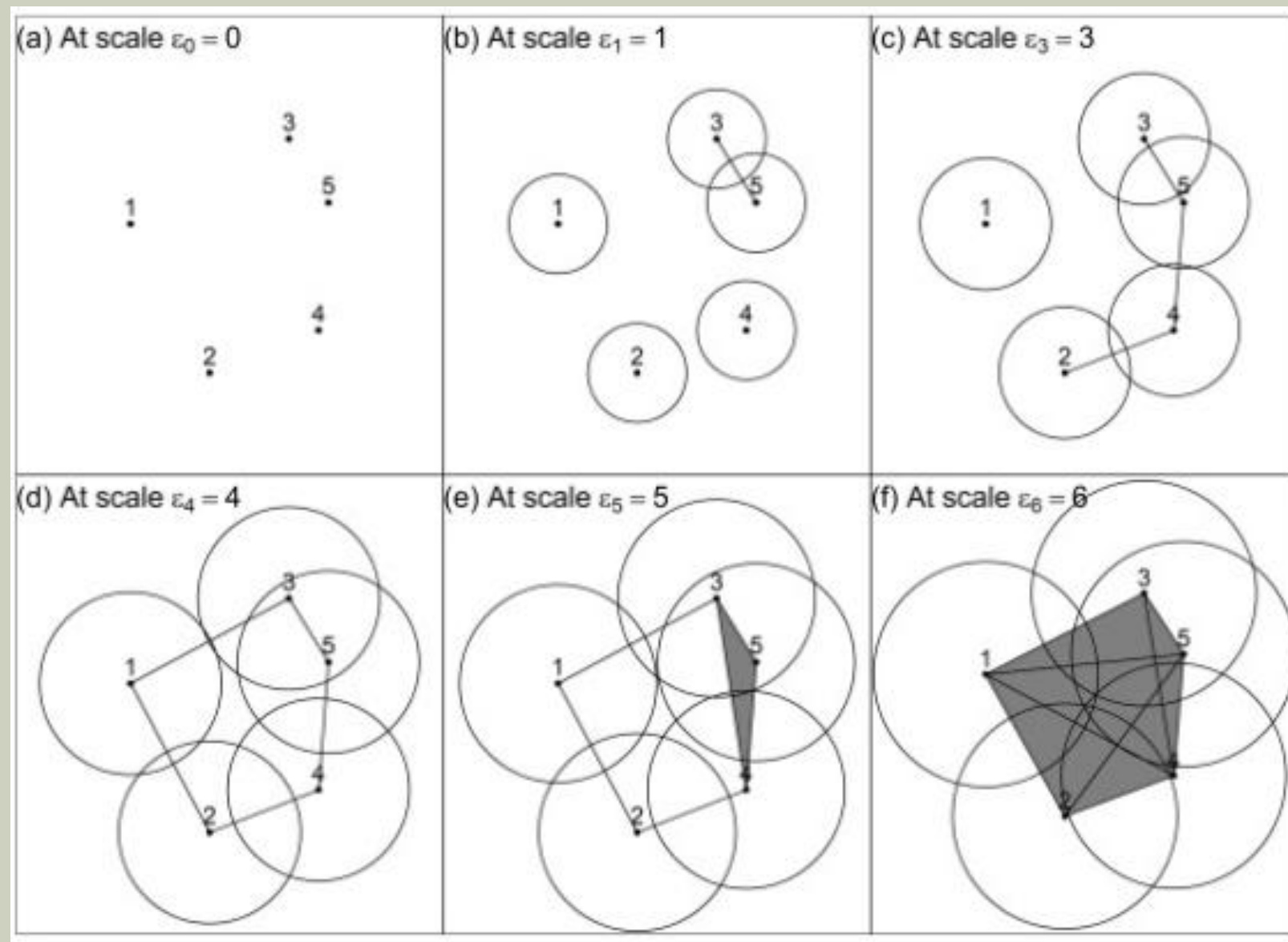
FINANCIAL CRISIS!



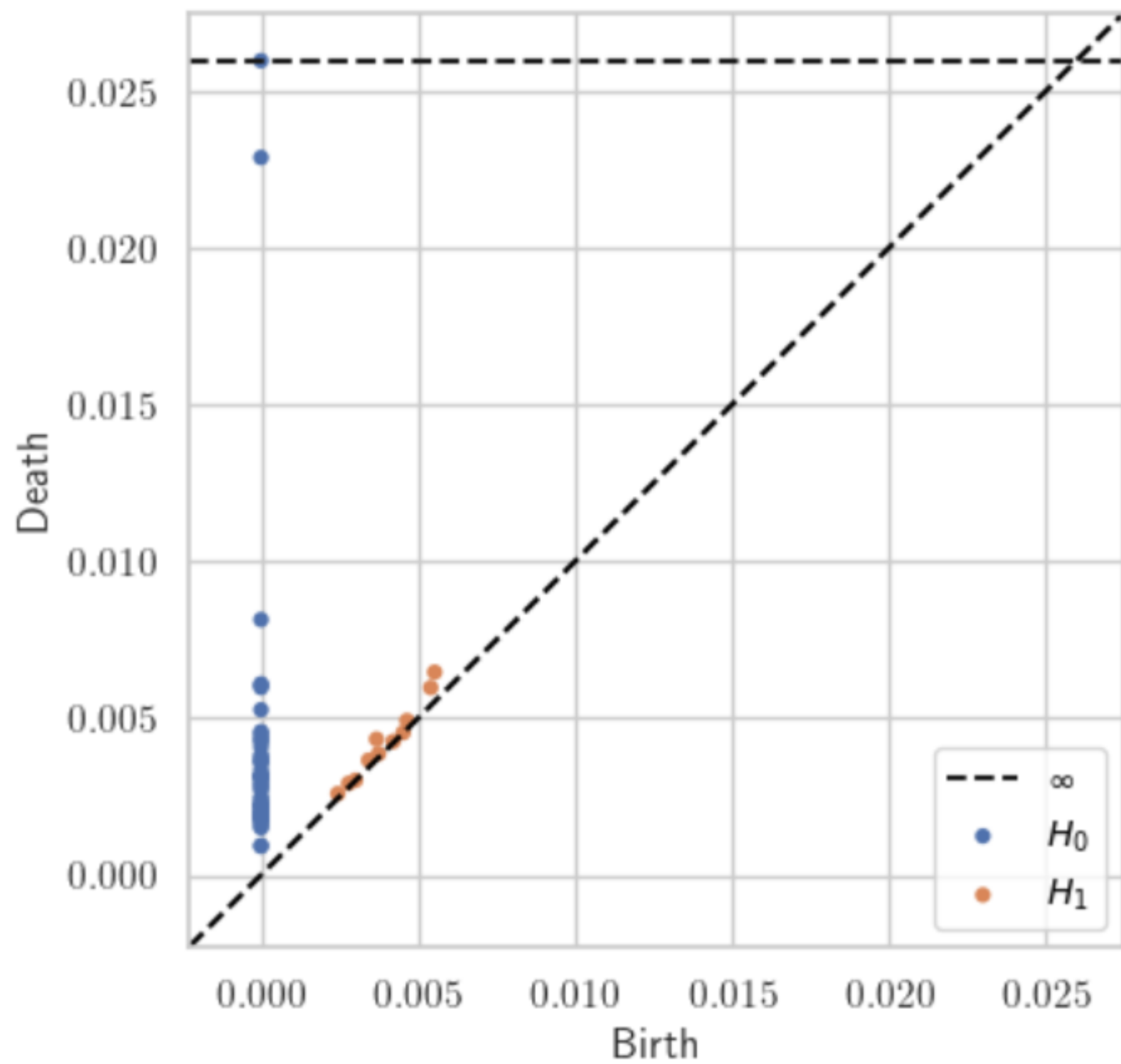
- Early Warning Signs are an active area of research!

2020 stock market crash	24 Feb 2020		The S&P 500 index dropped 34%, 1145 points, at its peak of 3386 on February 19 to 2237 on March 23. This crash was part of a worldwide recession caused by the COVID-19 lockdowns .
2018 cryptocurrency crash	20 Sep 2018		The S&P 500 index peaked at 2,930 on its September 20 close and dropped 19.73% to 2,351 by Christmas Eve. Bitcoin price peaked on 17 Dec '17, then fell 45% on 22nd Dec '17. The DJIA falls 18.78% during roughly the same period. Shanghai Composite dropped to a four-year low, escalating their economic downturn since the 2015 recession.
Financial crisis of 2007–2008	16 Sep 2008		On September 16, 2008, failures of large financial institutions in the United States, due primarily to exposure of securities of packaged subprime loans and credit default swaps issued to insure these loans and their issuers, rapidly devolved into a global crisis resulting in a number of bank failures in Europe and sharp reductions in the value of equities (stock) and commodities worldwide. The failure of banks in Iceland resulted in a devaluation of the Icelandic króna and threatened the government with bankruptcy. Iceland was able to secure an emergency loan from the IMF in November. Later on, U.S. President George W. Bush signs the Emergency Economic Stabilization Act into law, creating a Troubled Asset Relief Program (TARP) to purchase failing bank assets . Had disastrous effects on the world economy along with world trade.

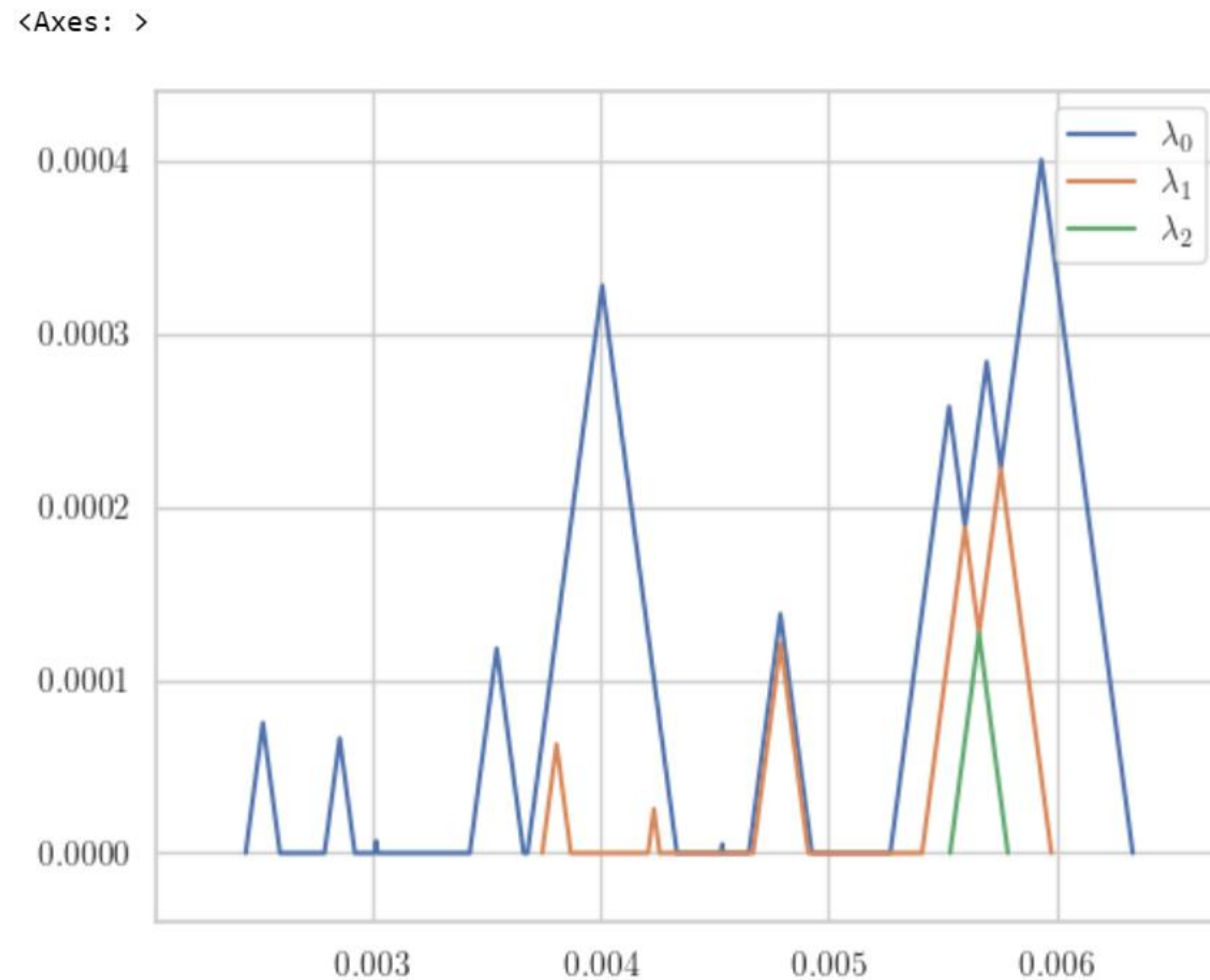
INTRODUCTION



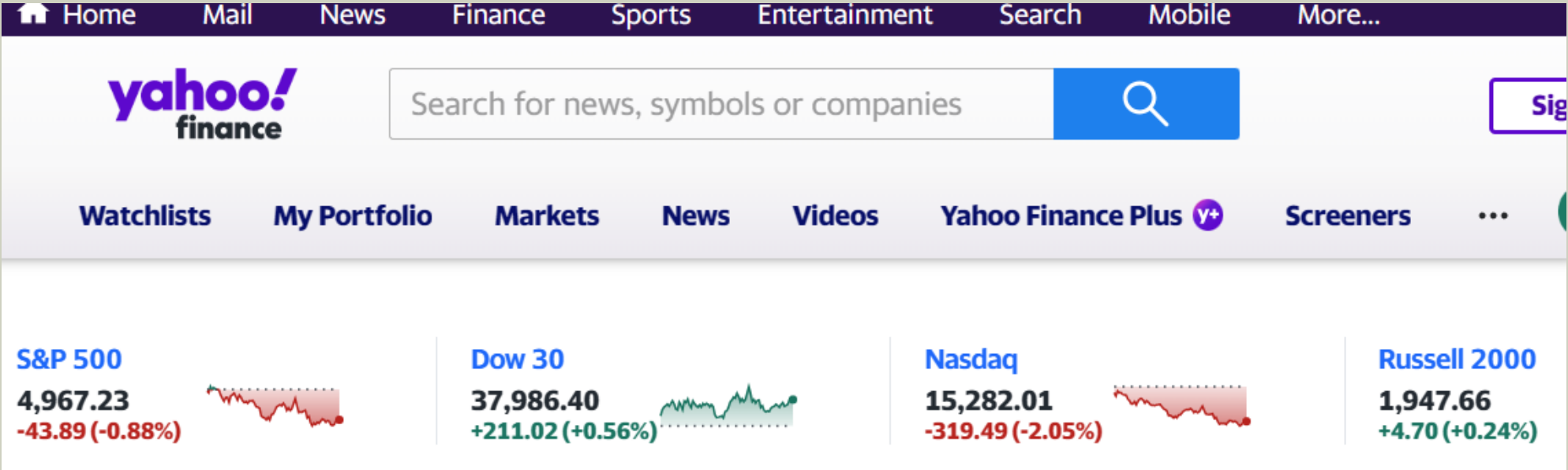
Persistence Diagram



Persistence Landscape



DATA

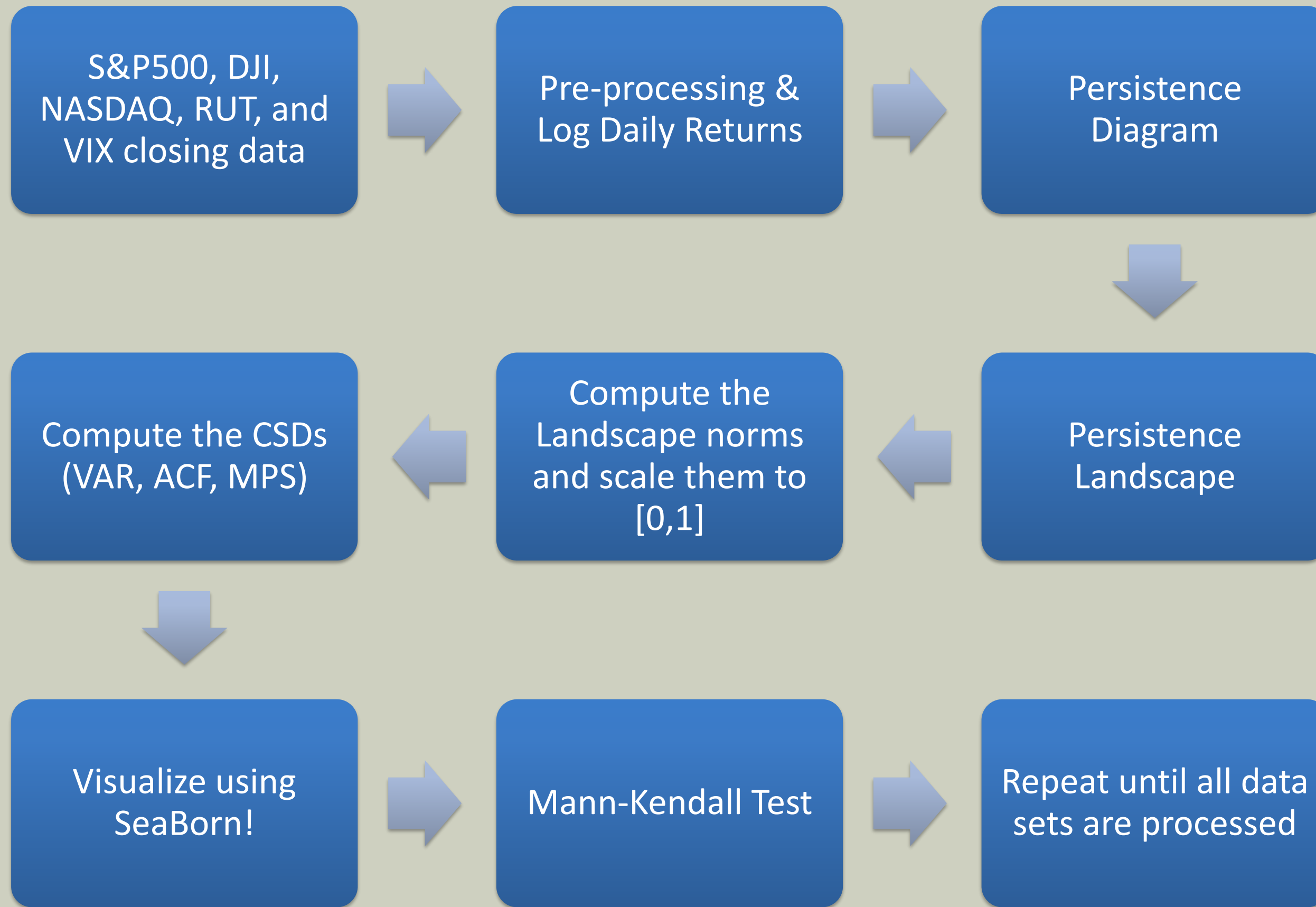


	A	B	C	D	E	F	G	H	
1	Date	Open	High	Low	Close	Adj Close	Volume		
2	1/2/2015	2058.9	2072.36	2046.04	2058.2	2058.2	2.71E+09		
3	1/5/2015	2054.44	2054.44	2017.34	2020.58	2020.58	3.8E+09		
4	1/6/2015	2022.15	2030.25	1992.44	2002.61	2002.61	4.46E+09		
5	1/7/2015	2005.55	2029.61	2005.55	2025.9	2025.9	3.81E+09		
6	1/8/2015	2030.61	2064.08	2030.61	2062.14	2062.14	3.93E+09		
7	1/9/2015	2063.45	2064.43	2038.33	2044.81	2044.81	3.36E+09		
8	1/12/2015	2046.13	2049.3	2022.58	2028.26	2028.26	3.46E+09		
9	1/13/2015	2031.58	2056.93	2008.25	2023.03	2023.03	4.11E+09		
10	1/14/2015	2018.4	2018.4	1988.44	2011.27	2011.27	4.38E+09		
11	1/15/2015	2013.75	2021.35	1991.47	1992.67	1992.67	4.28E+09		
12	1/16/2015	1992.25	2020.46	1988.12	2019.42	2019.42	4.06E+09		
13	1/20/2015	2020.76	2028.94	2004.49	2022.55	2022.55	3.94E+09		
14	1/21/2015	2020.19	2038.29	2012.04	2032.12	2032.12	3.73E+09		
15	1/22/2015	2034.3	2064.62	2026.38	2063.15	2063.15	4.18E+09		
16	1/23/2015	2062.98	2062.98	2050.54	2051.82	2051.82	3.57E+09		
17	1/26/2015	2050.42	2057.62	2040.97	2057.09	2057.09	3.47E+09		
18	1/27/2015	2047.86	2047.86	2019.91	2029.55	2029.55	3.33E+09		
19	1/28/2015	2032.34	2042.49	2001.49	2002.16	2002.16	4.07E+09		
20	1/29/2015	2002.45	2024.64	1989.18	2021.25	2021.25	4.13E+09		
21	1/30/2015	2019.35	2023.32	1993.38	1994.99	1994.99	4.57E+09		
22	2/2/2015	1996.67	2021.66	1980.9	2020.85	2020.85	4.01E+09		
23	2/3/2015	2022.71	2050.3	2022.71	2050.03	2050.03	4.62E+09		
24	2/4/2015	2048.86	2054.74	2036.72	2041.51	2041.51	4.14E+09		
25	2/5/2015	2043.45	2063.55	2043.45	2062.52	2062.52	3.82E+09		

Stock Indices

- Standard and Poor’s 500 (S&P 500)
- Dow Jones Industrial Average (DJIA)
- Nasdaq Composite (NASDAQ)
- Russel 2000 (RUT)
- Chicago Board Options Exchange Volatility Index (VIX)

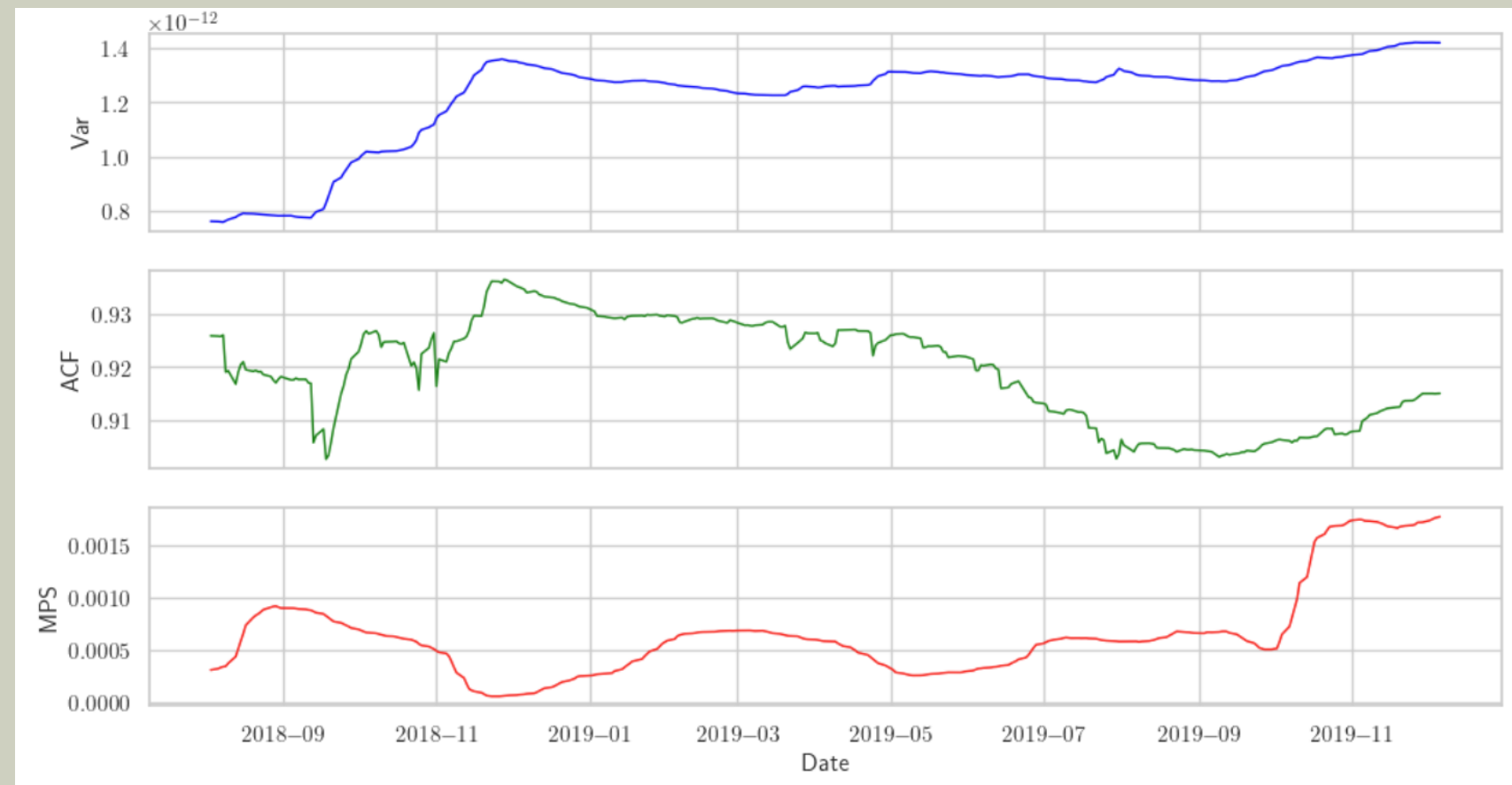
OUTLINE OF METHOD



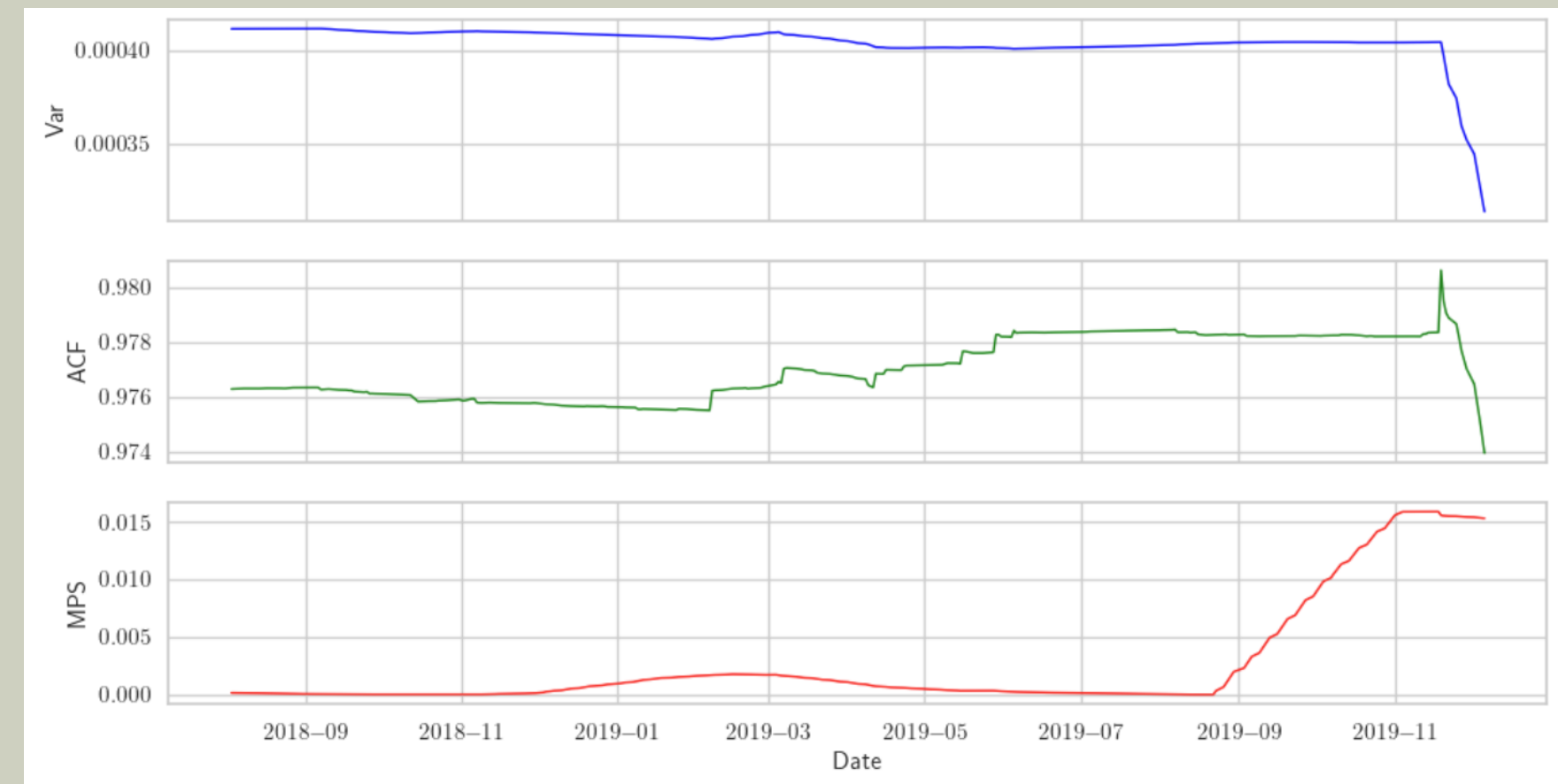
RESULTS

COMPARING STOCK MARKET CRASH 2020

CSD of Stock Market Crash
under 4-stock-aggregate with L1

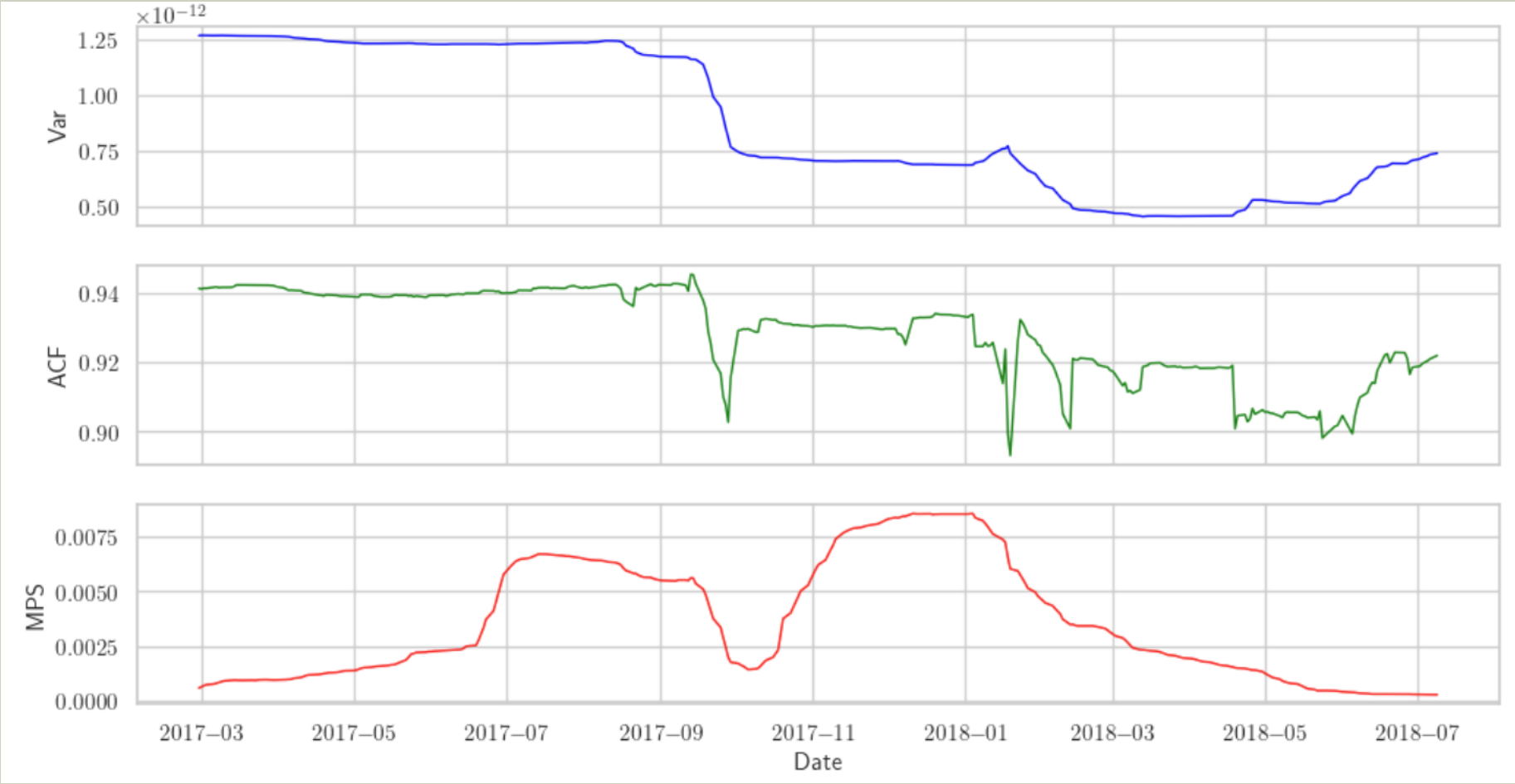


CSD of Stock Market Crash
under VIX with L1

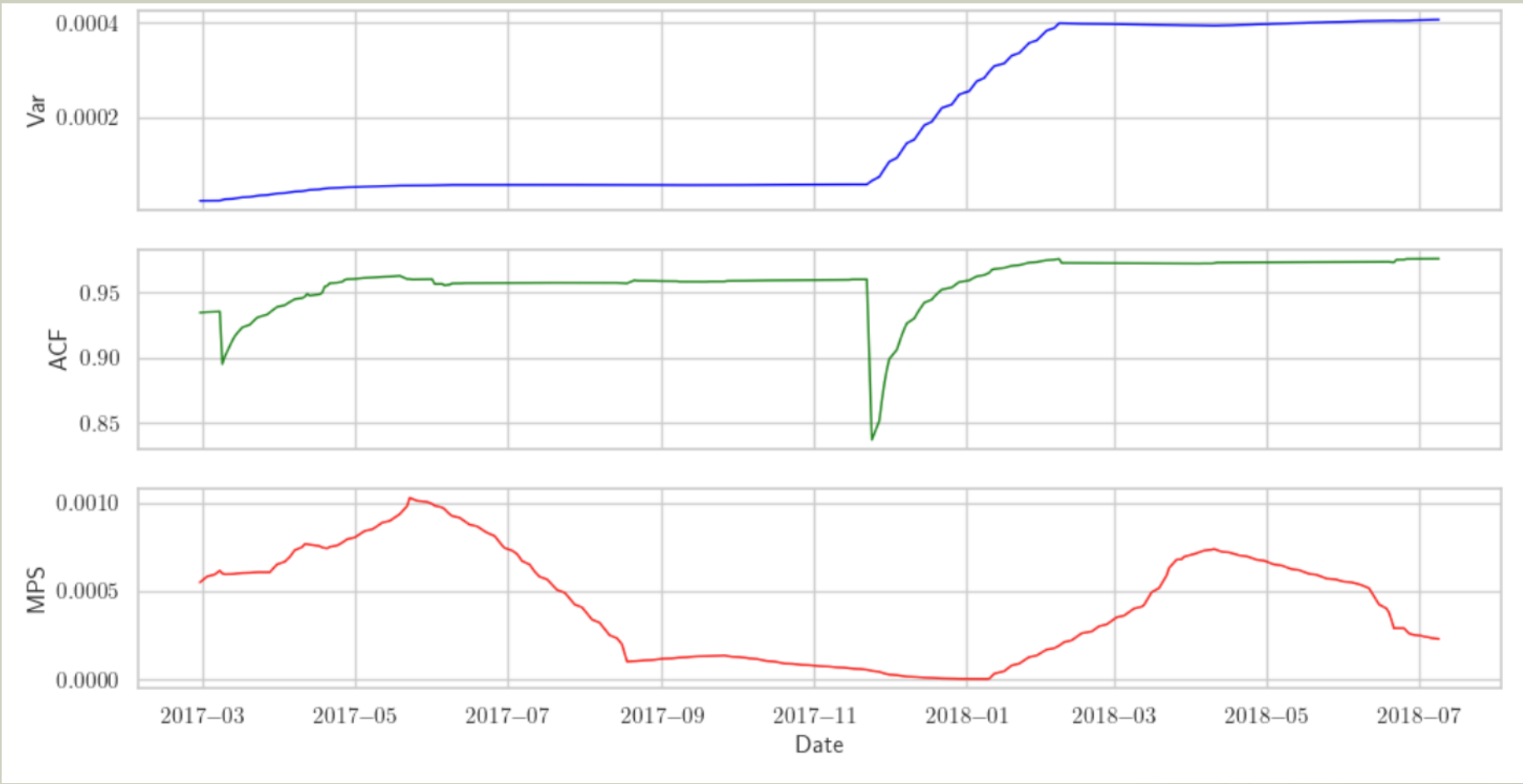


COMPARING CRYPTOCRASH 2018

CSD of Cryptocurrency Crash
under 4-stock-aggregate with L1

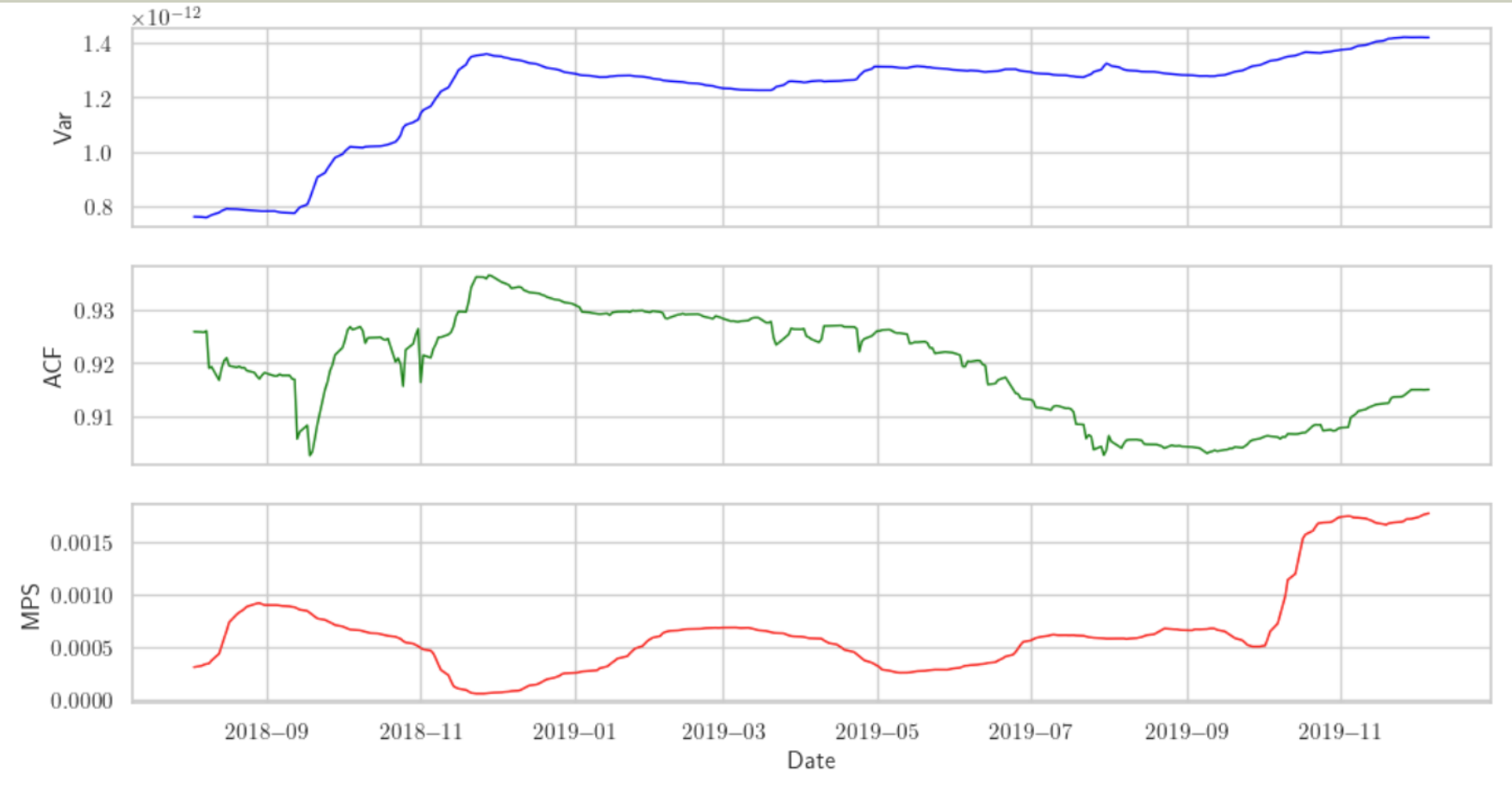


CSD of Cryptocurrency Crash
under VIX with L1

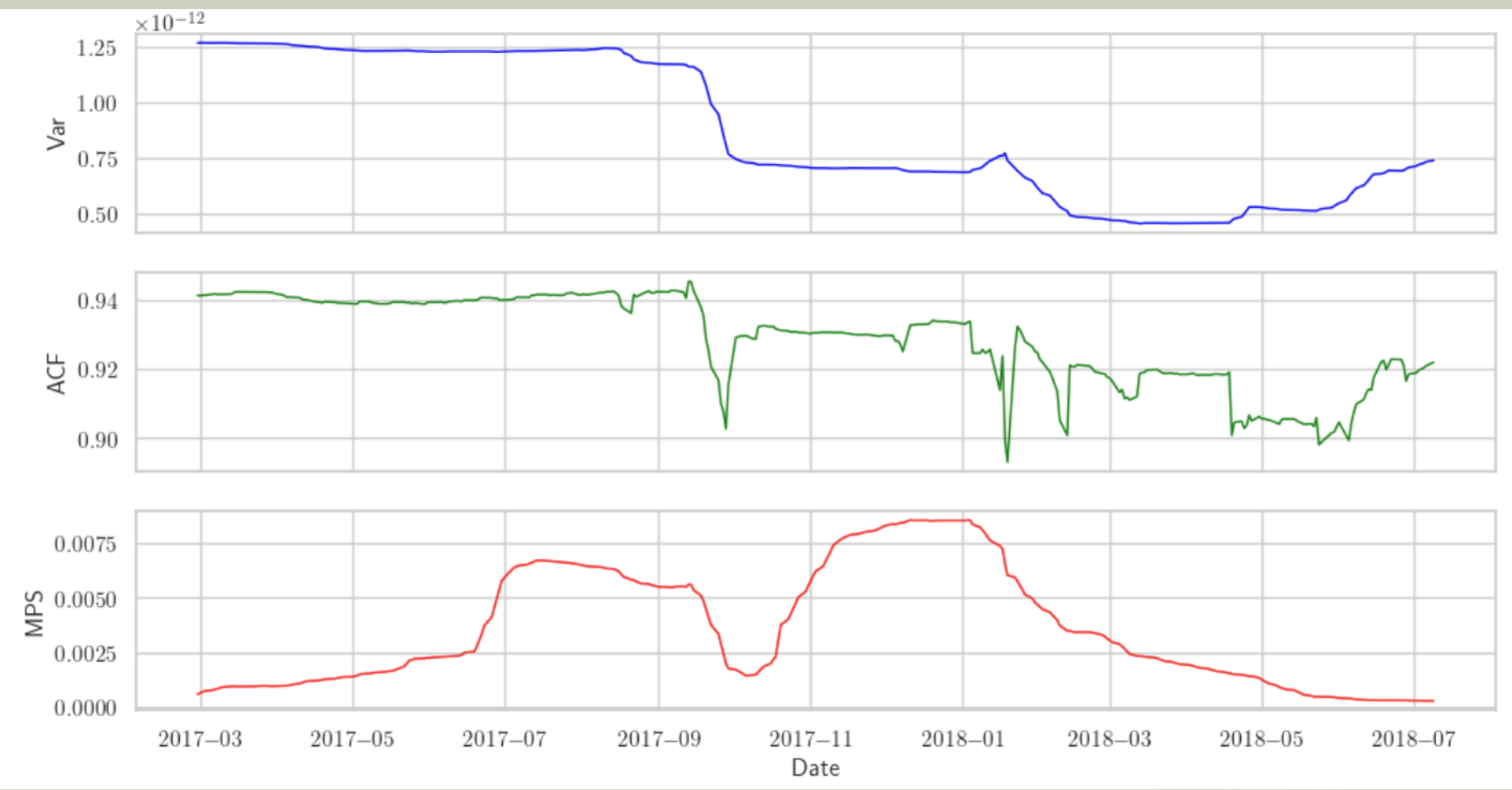


COMPARING 4-STOCK-AGGREGATE

CSD of Stock Market Crash
under 4-stock-aggregate with L1

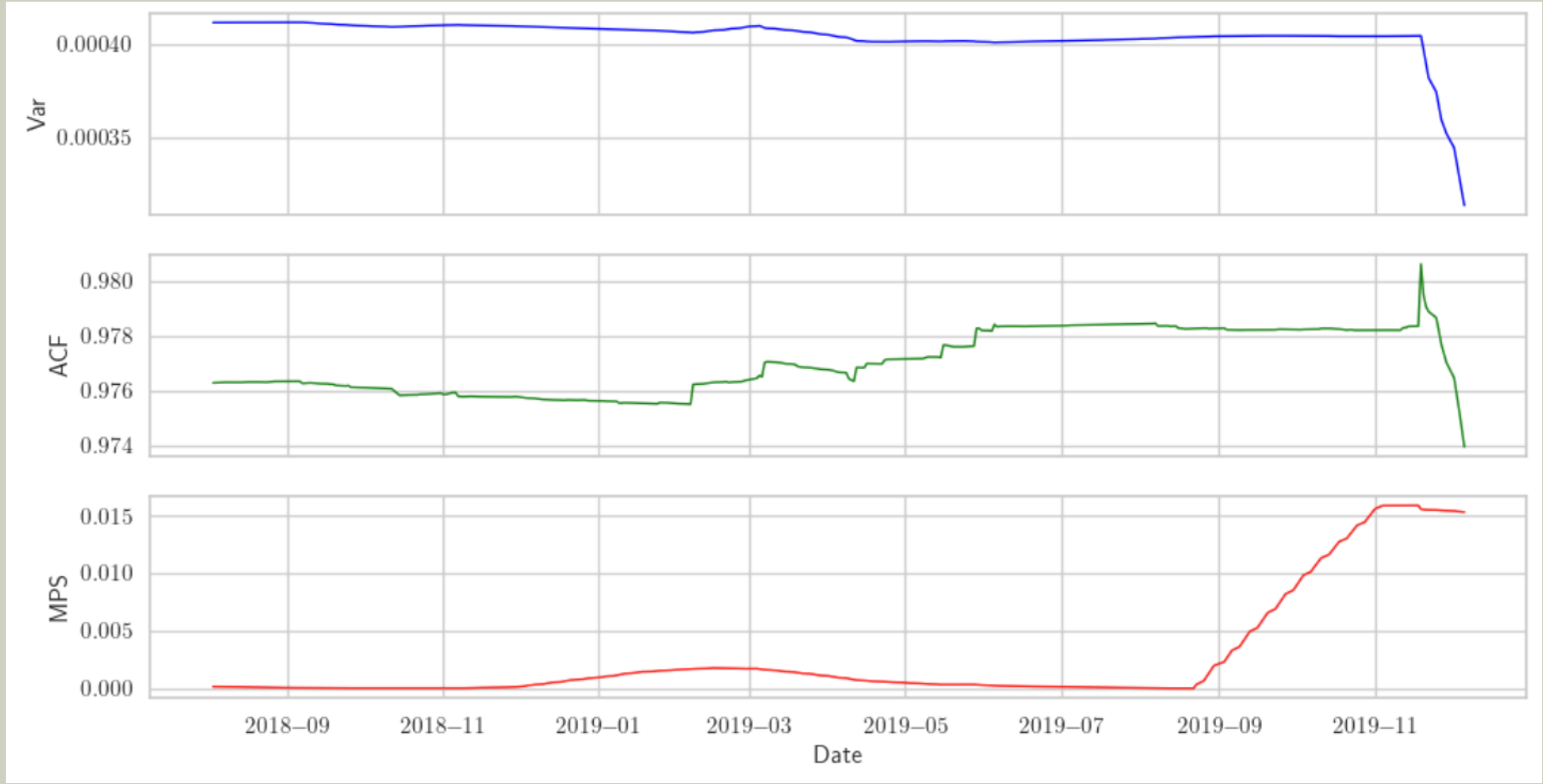


CSD of Cryptocurrency Crash
under 4-stock-aggregate with L1

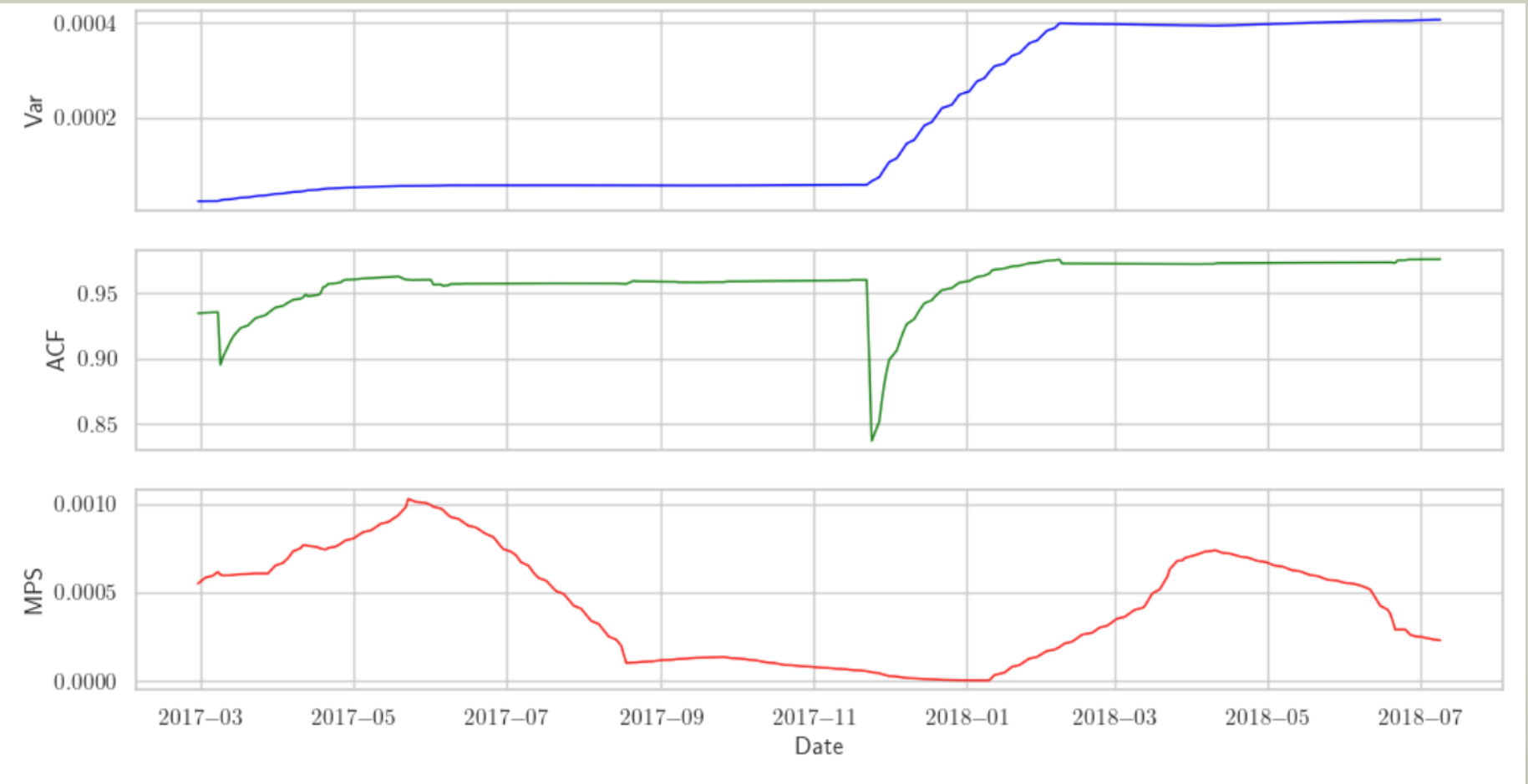


COMPARING VIX

CSD of Stock Market Crash
under VIX with L1



CSD of Cryptocurrency Crash
under VIX with L1



MANN-KENDALL TEST

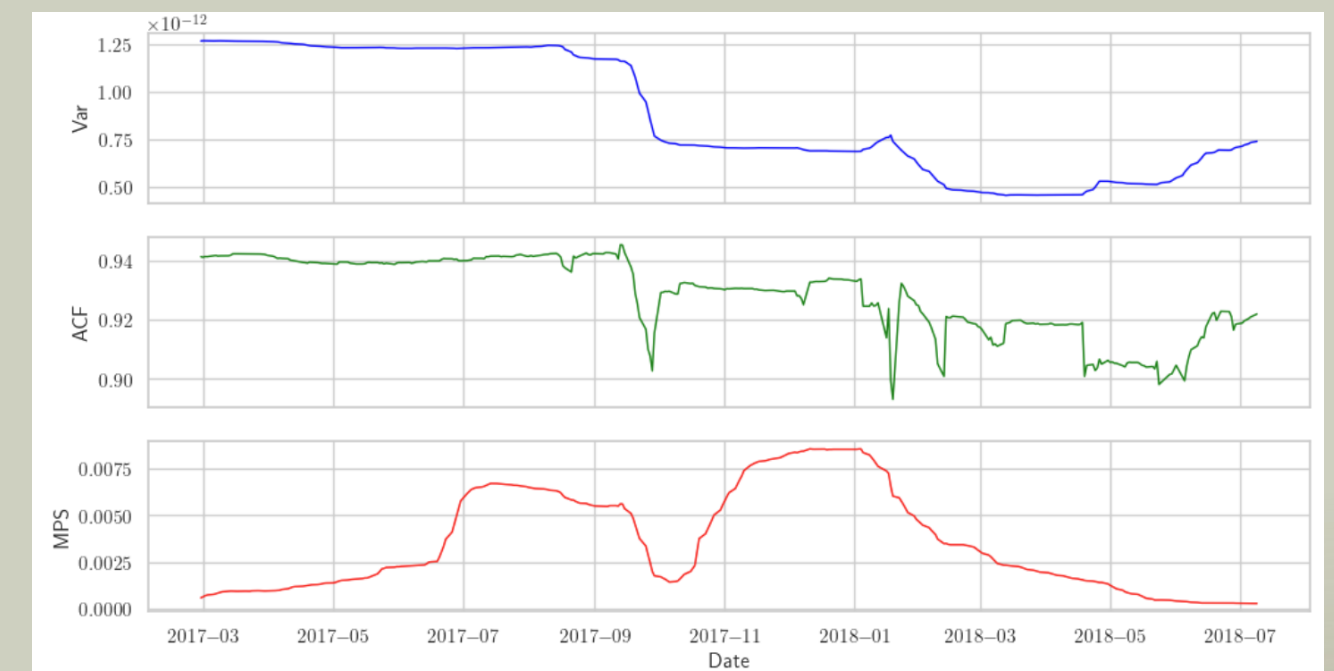
```
#MK Test for MPS in L2
print(mk.original_test(CSD_stock_L2['MPS']))
print(mk.original_test(CSD_crypto_L2['MPS']))
print(mk.original_test(CSD_VIX_stock_L2['MPS']))
print(mk.original_test(CSD_VIX_crypto_L2['MPS']))
```

```
Mann_Kendall_Test(trend='increasing', h=True, p=4.1569876430003205e-09, z=
5.877822320823899, Tau=0.21394285315320033, s=12257.0, var_s=4347750.333333
333, slope=1.6398510695135514e-06, intercept=0.0005627471085344004)
Mann_Kendall_Test(trend='no trend', h=False, p=0.5216013169029956, z=-0.640
8789750769072, Tau=-0.023204269176342215, s=-1361.0, var_s=4503247.0, slope
=-8.796786620392116e-07, intercept=0.0028128362050691903)
Mann_Kendall_Test(trend='increasing', h=True, p=0.0, z=9.771112268641165, T
au=0.3556405019985687, s=20375.0, var_s=4347750.333333333, slope=1.14798231
23153945e-05, intercept=-0.0012496832496654117)
Mann_Kendall_Test(trend='decreasing', h=True, p=0.0005676686129205688, z=-
3.446609429200367, Tau=-0.12471655328798185, s=-7315.0, var_s=4503247.0, sl
ope=-8.309205983296099e-07, intercept=0.0007657645958795719)
```



Results of Mann-Kendall Test						
	L1			L2		
Name	VAR	ACF	MPS	VAR	ACF	MPS
Stock2020	Increasing	Decreasing	Increasing	Increasing	Decreasing	Increasing
Crypto2018	Decreasing	Decreasing	Decreasing	Decreasing	Decreasing	No trend
VIX2020	Decreasing	Increasing	Increasing	Decreasing	Increasing	Increasing
VIX2018	Increasing	Increasing	Decreasing	Increasing	Increasing	Decreasing
p-values from Mann-Kendall Test						
	L1			L2		
Name	VAR	ACF	MPS	VAR	ACF	MPS
Stock2020	0	0	0	0	0	4.157
Crypto2018	0	0	0.045	0	0	0.522
VIX2020	0	0	0	0	0	0
VIX2018	0	0	0	0	0	0.001

CSD of Cryptocurrency Crash
under 4-stock-aggregate with L1



INCONCLUSIVE?

FUTURE RESEARCH

IMPACT OF THE
SLIDING WINDOW

ABILITY TO
DETECT LOCAL
MARKET CRASHES

INTERPRETATION OF
LOOPS

PRECISION/ACCURACY